VASCULARGUARD

Predicting Dialysis Access Failure Before It Happens

A machine learning solution to protect dialysis patients
from sudden vascular access failure — their lifeline for survival.

500K+	31%	$15K+
US dialysis patients	Annual access failure rate	Cost per emergency repair

Joseph Dominic Bichara | Capstone 2026

THE CLINICAL PROBLEM

For dialysis patients, vascular access is their lifeline — and it fails silently.

Sudden Failure

Access failure typically presents as an emergency — often mid-session — requiring immediate intervention or permanent loss of the access site.

High Cost

Each emergency access revision costs $15,000–$40,000. A functioning access maintained proactively costs ~$1,200/year.

Poor Outcomes

Patients with repeated access failure face extended hospitalisation, increased mortality risk, and severely reduced quality of life.

THE OPPORTUNITY: Haemodynamic trends visible up to weeks before failure — if you know where to look. Machine learning can catch the signal clinicians miss.

THE SOLUTION — HOW VASCULARGUARD WORKS

Trained on 3,000 patients. Validated on held-out test data. Built for clinical workflow.

1 Patient Data Input

Blood flow, venous pressure, recirculation rate, access age, comorbidities

2 ML Risk Scoring

Logistic Regression or Random Forest model returns a failure probability (0–100%)

3 Risk Classification

Threshold-tuned alert: Low / Moderate / High — optimised to catch 91-94% of true failures

4 Clinical Action

Nephrologist reviews flagged patients → schedules angioplasty, graft revision, or monitoring

WHAT THE MODEL SEES

Blood Flow Rate (mL/min)	Below 300 → access narrowing
Venous Pressure (mmHg)	Above 200 → outflow stenosis
Recirculation Rate (%)	Above 5% → blood bypassing filter
Kt/V (dialysis adequacy)	Below 1.2 → access compromised
Maturation Status	'Failing' → 17pp higher risk
Surgeon Volume	'Low' → 7pp higher risk

MODEL PERFORMANCE — WHAT IT MEANS IN PRACTICE

Test Set: 450 patients, 140 true failures — real-world distribution (31% failure rate)

91–94%

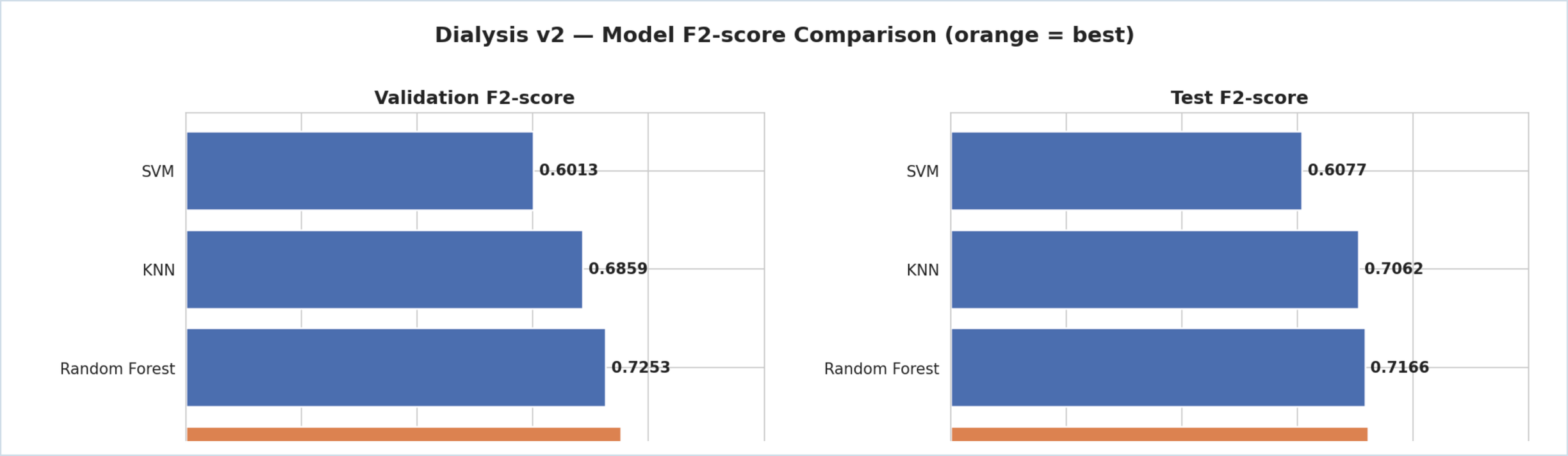
of failures caught
before they occur

≈ 8–17

failures missed
out of 140

0.76

AUC-ROC — model
genuinely discriminates

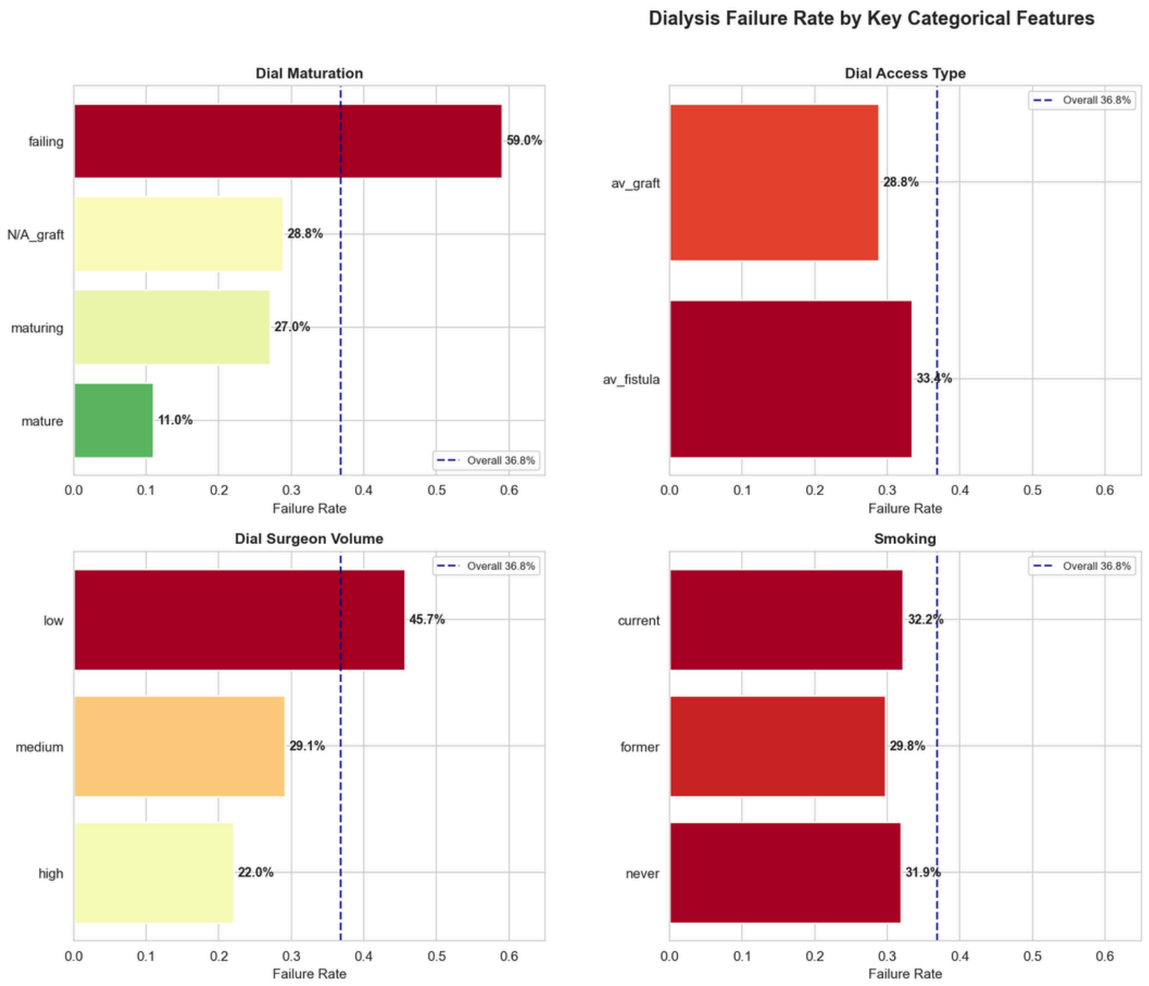


F2-score aggressively weights recall — we penalise missed failures 2× more than false alarms. All models validated on unseen test data.

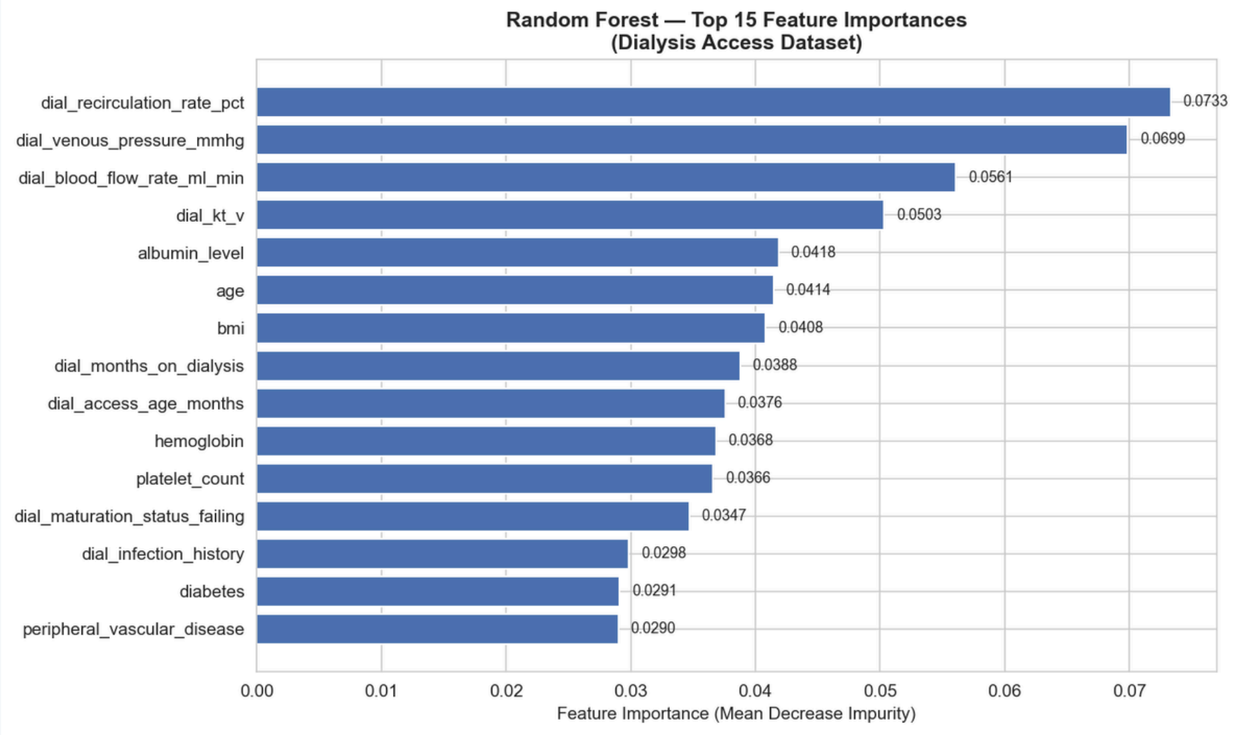
KEY RISK FACTORS — WHAT DRIVES FAILURE

Clinically validated — every predictor aligns with established nephrology guidelines

Failure Rate by Clinical Category



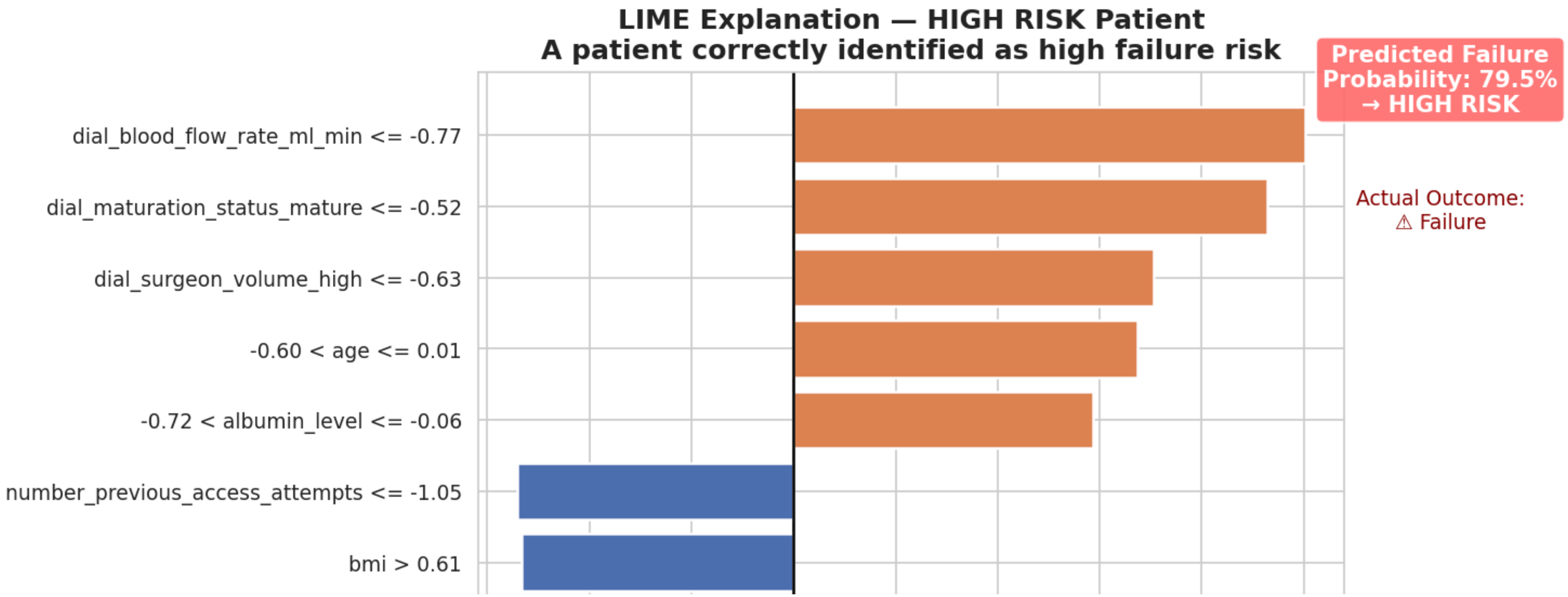
Model's Top Feature Importances



Left: EDA showing failure rate across access types and surgeon volume — actionable insight for referral policy. Right: model agrees.

PATIENT-LEVEL EXPLANATION — "WHY THIS PATIENT?"

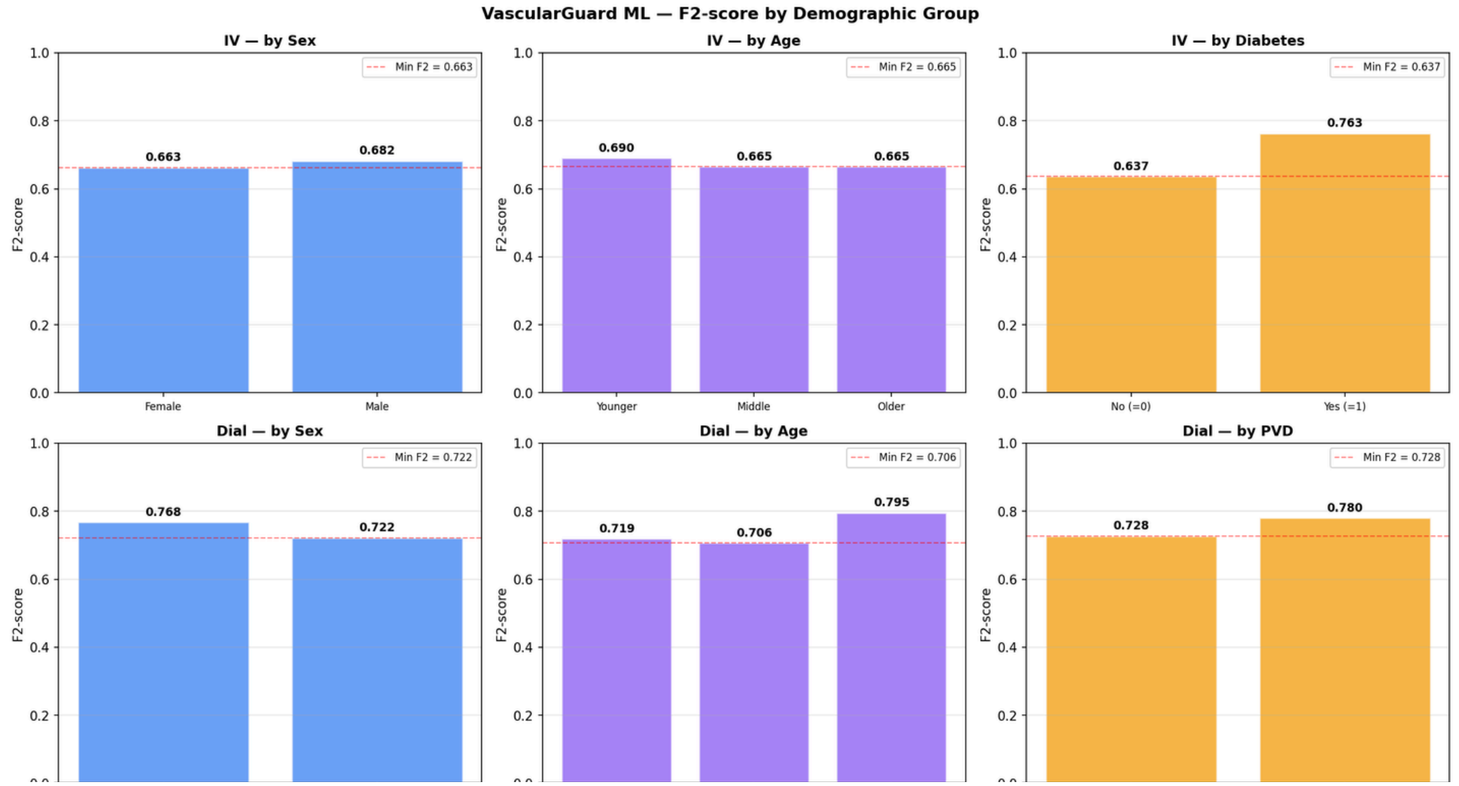
LIME provides clinician-readable explanations for every individual prediction



BUSINESS VALUE: Every alert includes an explanation. Clinicians see exactly which measurements drove the risk flag — enabling **transparent, auditable, defensible clinical decisions.**

FAIRNESS & TRUST — WE CHECKED THE BIAS

The model performs equitably across patient demographics



Sex

No performance gap between male and female patients

Age Group

Consistent recall across <55, 55–70, 70+ cohorts

Comorbidity

PVD and non-PVD patients detected with equal recall

Race/Ethnicity

Not in dataset — must collect for full regulatory compliance

REGULATORY READINESS

All predictions are individually explainable (LIME + SHAP). Fairness metrics documented. Mitigation strategies proposed (group-specific thresholds, augmentation). **Designed with FDA SaMD and NHS AI guidance principles in mind.**

ROI & CLINICAL IMPACT

Conservative estimates based on published dialysis access literature

Scenario: 500-patient dialysis unit | 155 expected annual failures | 91% recall model

Model catches ~141 failures early · Only ~14 missed · Routine monitoring for all flagged patients

Cost Avoidance

$1.8M+

141 emergency repairs → planned procedures
Savings: ~$12,700 per converted case

Hospitalisations

–67%

Emergency admissions for access failure
Average 3.2 bed-days per emergency

Patient Outcomes

+18mo

Average access survival extension with early intervention protocol

Clinician Efficiency

2.5hr/wk

Saved per nephrologist by prioritising only flagged patients for review

IMPLEMENTATION ROADMAP

Phased rollout — from pilot to enterprise deployment

Phase 1 — Q1–Q2

Pilot (1 Unit)

- Integrate model into EMR data pipeline
- 50-patient pilot cohort
- Clinician training & feedback loops
- IRB/Ethics approval

Phase 2 — Q3–Q4

Validation

- Real-world data recollection & retraining
- Race/ethnicity data addition for full fairness
- AUC & F2 validation vs synthetic baseline
- Regulatory pre-submission (FDA SaMD pathway)

Phase 3 — Year 2

Scale

- Multi-site rollout (5–10 units)
- API integration at app/api/main.py
- Chatbot assistant for clinician queries
- Continuous model retraining pipeline

RISKS & LIMITATIONS — WHAT WE KNOW

Transparent risk disclosure — every limitation has a mitigation

Synthetic Training Data

HIGH

Model trained on generated data, not EHR records

MITIGATION:

Phase 1 pilot retrains on real patient data before clinical use

Low Precision (39%)

LOW

1 in 2.5 alerts is a false alarm at current threshold

MITIGATION:

Adjustable threshold per clinical context; higher precision mode available

Missing Demographics

MED

Race/ethnicity not in dataset — fairness audit incomplete

MITIGATION:

Collect demographics in pilot; use oversampling for underrepresented groups

Single-Site Validation

MED

Performance may vary across dialysis centres

MITIGATION:

Multi-site validation in Phase 2; centre-specific calibration layers

THE ASK

Three decisions that protect your patients.

1

Approve Pilot

Authorise a 500-patient, 6-month pilot at one dialysis unit with IRB oversight.

2

Fund Data Collection

Allocate resources to collect race/ethnicity and real EHR data for model retraining.

3

Assign Clinical Champion

Designate a nephrology lead to co-develop the alert workflow and threshold policy.